20.5 Local bond behaviour of steel reinforcement

20.5.1 Local bond-slip relationship for ribbed bars for predominantly static loading in case of no degradation

20.5.1.1 Local bond stress–slip model

Under well-defined conditions, it is possible to formulate an average "local bond" versus "local slip" relationship, for short anchorage lengths, for instance up 5 times bar diameter.
This section covers local bond behaviour of ribbed reinforcing bars which satisfy the requirements for classification as "high bond" in accordance with requirements of chapter 15.

For monotonic loading the reference value τ_b of the bond stresses between concrete and reinforcing bar for pull-out and splitting failure can be calculated as a function of the relative displacement parallel to the bar axis "s" as follows (Fig. 20.5-1):

$$\tau_b = \tau_{bmax} (s/s_1)^{\alpha} \quad \text{for} \quad 0 \le s \le s_1 \qquad (20.5\text{-}1)$$

$$\tau_b = \tau_{bmax} \quad \text{for} \quad s_1 \le s \le s_2 \qquad (20.5\text{-}2)$$

$$\tau_b = \tau_{bmax} - (\tau_{bmax} - \tau_{bf}) (s\text{-}s_2)/(s_3\text{-}s_2) \quad \text{for} \quad s_2 \le s \le s_3 \qquad (20.5\text{-}3)$$

$$\tau_b = \tau_{bf} \quad \text{for} \quad s_3 < s \qquad (20.5\text{-}4)$$

where the parameters are given in Table 20.5-1.

The bond stress–slip relationship depends on a considerable number of influencing factors including rib geometry (relative rib area), concrete strength, position and orientation of the bar during casting, state of stress, boundary conditions and concrete cover. The parameters given in Table 20.5-1 are valid for ribbed reinforcing steel with a relative rib area $f_r \ge 0.05\, f_{r,min}$ according to relevant international standards.
The bond stress–slip curves for confined and unconfined concrete presented in Fig. 20.5-1 can be considered applicable as an average formulation for a broad range of cases. Further reliability handling would be needed to derive design bond stress–slip curves.
The ascending part refers to the stage in which the ribs penetrate into the mortar matrix, characterized by local crushing and micro-cracking. A sustained plateau occurs only for confined concrete, during which advanced crushing and shearing off of the concrete between the ribs takes place. This level represents a residual bond capacity which is maintained only where a large concrete cover, dense transverse reinforcement or transverse compression is present to keep a certain degree of integrity intact. The descending branch refers to the reduction of bond resistance as concrete corbels between the ribs are sheared off. In the case of unconfined concrete, splitting failure occurs which is reflected by a sudden drop in bond stress before a constant residual level is reached. The peak value of bond strength in a splitting failure mode is denoted $\tau_{bu,split}$ – see Fig. 20.5-1 and Eq. (20.5-5).
With regard to the development of bond stresses, the following considerations apply:
reinforcement and concrete have the same strain ($\varepsilon_s = \varepsilon_c$) in those areas of the structure where the steel is in compression (outside eventual load introduction

areas) and in those areas where the steel is in tension in uncracked parts of the structure.

In cracked cross-sections, tension forces are transferred across the crack by the reinforcing steel. In general, the absolute displacements of the steel u_s and of the concrete u_c adjacent to a crack are different. Differences in displacements are similarly found along the transmission length l_{bpt} of pretensioned prestressed concrete members.

Due to the relative displacement $s = u_s - u_c$ bond stresses are generated between concrete and reinforcing steel or prestressing tendons. For $s < s_1$, the magnitude of these bond stresses depends predominantly on the slip s, but is influenced by the surface of the steel, concrete strength f_{cm}, the position of the reinforcing steel during concreting and by confinement from concrete cover, secondary reinforcement and the state of stress in the concrete around the bar. Between cracks or along the transmission length l_{bpt} a part of the tension force of the reinforcing steel at the cracked section is transferred into the concrete by bond (tension stiffening effect). The local change in relative displacement is characterized by the strain difference $ds/dx = \varepsilon_s - \varepsilon_c$.Depending on the selection of the coefficient α ($0 \le \alpha \le 1$) in Eq. (20.5-1) all usual forms of a bond stress–slip relationship can be modelled, starting from a bond characteristic with a constant stress ($\alpha = 0$) up to a bond stress–slip relationship with linear increasing bond stress ($\alpha = 1$).

The parameters from Table 20.5-1 to be applied are dependent primarily on the failure mode (pull-out or splitting). Further distinctions are made to account for differences in bond conditions within each failure mode.

The rationale underpinning the following rules is given in *fib* Bulletin 72 [20-1].

The values in Table 20.5-1, columns 3 to 6 (splitting failure) are derived from Eq. (20.5-5), which is derived from the semi-empirical formulation of bond presented in the *fib* Bulletin 72 [20-1] by setting $l_b/Ø = 5$, assuming a uniform bond stress over this length and evaluating $\tau_{bu,split}$ for Ø = 25 mm, $c_{max}/c_{min} = 2.0$, $c_{min} = Ø$ and $K_{tr} = 0.02$ in the case where bars are confined by stirrups or $K_{tr} = 0$ where no stirrups are provided.

$$\tau_{bu,split} = \eta_2 6.5 \left(\frac{f_{cm}}{25}\right)^{0.25} \left(\frac{25}{Ø}\right)^{0.2} \left[\left(\frac{c_{min}}{Ø}\right)^{0.25} \left(\frac{c_{max}}{c_{min}}\right)^{0.1} + k_m K_{tr}\right] \qquad (20.5\text{-}5)$$

where:

η_2 = 1.0 for good bond conditions;

= 0.7 for all other bond conditions (see section 20.2.2.1 for definition of bond conditions);

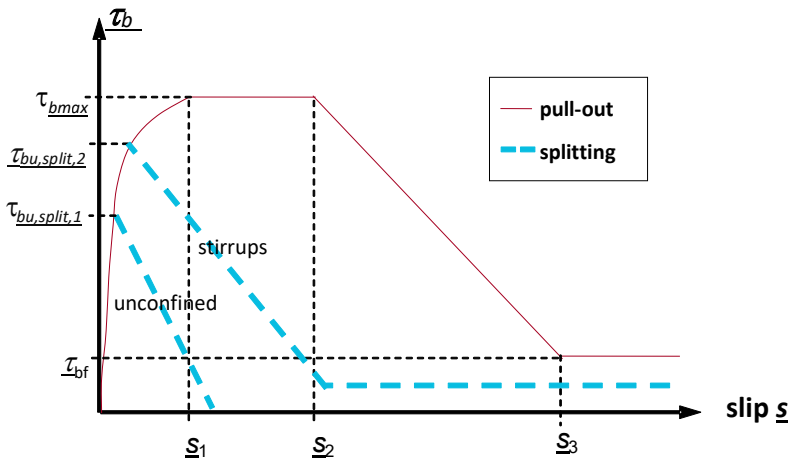


Figure 20.5-1: Analytical bond stress–slip relationship (monotonic loading). $\tau_{bu,split,1}$ and $\tau_{bu,split,2}$ denote peak local bond resistance in the absence ($K_{tr} = 0$) and presence, respectively, of confining stirrups. TO BE RE-DRAFT

f_{cm} is the mean cylinder concrete compressive strength [N/mm^2];

Ø is the diameter of the anchored bar considered [mm];

c_{min} = min{c_s/2, c_x, c_y};

c_{max} = max{c_s/2, c_x}.

k_m represents the efficiency of confinement from transverse reinforcement, and has a value of 12 where bars are confined inside a bend of links passing round the bar of at least 90° within a distance of 5ϕ (see Fig. 20.5-3). Where no confining reinforcement is provided between bars and the nearest face, k_m = 0, as shown in Fig. 20.5-3. Other values of k_m are described in Fig. 20.5-3. K_{tr} is defined in section 20.2.2.3.

Table 20.5-1: Parameters defining the mean bond stress–slip relationship of ribbed bars (according to Eqs. (20.5-1)–(20.5-4)).

	1	2	3	4	5	6
	Pull-out (PO)		Splitting (SP)			
	$\varepsilon_s < \varepsilon_{s,y}$		$\varepsilon_s < \varepsilon_{s,y}$			
	Good bond cond.	All other bond cond.	Good bond cond.		All other bond cond.	
			Unconfined	Stirrups	Unconfined	Stirrups
τ_{bmax}	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$2.5\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$	$1.25\sqrt{f_{cm}}$
$\tau_{bu,split}$	—	—	$7.0\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$	$8.0\cdot\left(\frac{f_{cm}}{25}\right)^{0}$	$5.0\cdot\left(\frac{f_{cm}}{25}\right)^{0.25}$	$5.5\cdot\left(\frac{f_{cm}}{25}\right)^{0.2}$
s_1	1.0 mm	1.8 mm	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$	$s(\tau_{bu,split})$
s_2	2.0 mm	3.6 mm	s_1	s_1	s_1	s_1
s_3	c_{clear} [1)]	c_{clear} [1)]	$1.2s_1$	$0.5c_{clear}$ [1)]	$1.2s_1$	$0.5c_{clear}$ [1)]
α	0.4	0.4	0.4	0.4	0.4	0.4
τ_{bf}	$0.40\,\tau_{max}$	$0.40\,\tau_{max}$	0	$0.4\,\tau_{bu,split}$	0	$0.4\,\tau_{bu,split}$

[1)] c_{clear} is the clear distance between ribs

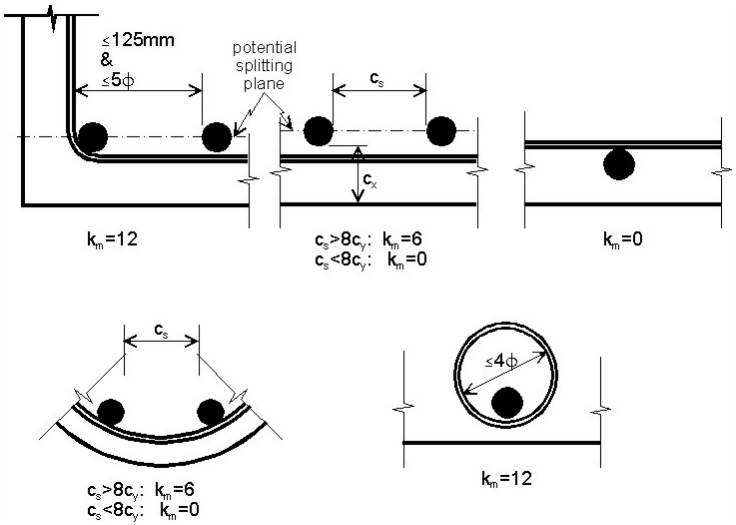


Figure 20.5-3: Confinement coefficients for transverse reinforcement

The values in Table 20.5-1, columns 1 and 2 (pull-out failure) are valid for well-confined concrete (concrete cover ≥ 5Ø, clear spacing between bars ≥ 10 Ø), or suitable confining reinforcement.

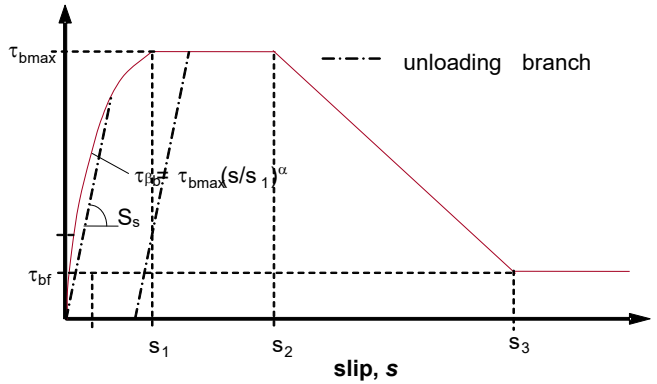


Figure 20.5-2: Unloading branch of the τ_b–s relationship.

The slope S_s of the unloading branch is taken as the secant modulus at a slip of 0.01 mm.
Unloading from a maximum slip of less than 0.01 mm is taken back to the origin, that is to zero slip.

The unloading branch of the bond stress–slip relationship is linear and valid for the increasing and horizontal part of the diagram. The slope S_s (see Fig. 20.5-2) is independent of the slip value s, and is given by:

$$S_s = 6.0\ \tau_{bmax}\ [\text{N/mm}^3] \qquad (20.5\text{-}6)$$

20.5.1.2 Influence of transverse cracking

For those parts of the reinforcing bar which are at a distance x ≤ 2 Ø from a transverse crack, the bond stress τ_B has to be reduced by the factor λ where:

$$\lambda = 0.5\, x/Ø \leq 1 \qquad (20.5\text{-}7)$$

20.5.1.3 Influence of yielding, transverse stress, longitudinal cracking and cyclic loading

The bond stress τ_b according to Eqs. (20.5-1)–(20.5-4) should be modified by the factors Ω_y, $\Omega_{p,tr}$ and Ω_{cr} and Ω_{cyc} in case of bar yielding, transverse pressure and cracking parallel to the bar axis and cyclic loading respectively:

$$\tau_{b,md} = \tau_{\mathrm{bm}}\, \Omega_{\mathrm{y}}\, \Omega_{\mathrm{p,tr}}\, \Omega_{\mathrm{cr}}\, \Omega_{\mathrm{cyc}} \qquad (20.5\text{-}8)$$

where:

$\tau_{\mathrm{b,md}}$ is bond stress according to the modified bond stress–slip curve;

τ_{b} is bond stress according to the bond stress–slip curve defined by Eqs. (20.5-1)–(20.5-4).

Ω_y is a factor which takes into account the influence of yielding of reinforcement

$\Omega_{p,tr}$ is a factor which takes into account the influence of transverse pressure

Ω_{cr} is a factor which takes into account the influence of longitudinal cracking

Ω_{cyc} is a factor which takes into account the influence of reversed cyclic loading

The modified bond stress $\tau_{\mathrm{b,md}}$ then replaces τ_{b} determined according to Eqs. (20.5-1)–(20.5-4).

20.5.1.3.1 Influence of yielding of reinforcement

If yielding of reinforcement occurs along the embedment length, the corresponding reduction of the bond stress is given by the factor Ω_y as:

$$\Omega_y = 1.0 \qquad \text{for } \varepsilon_s \leq \varepsilon_{sy} \qquad (20.5\text{-}9a)$$

$$\Omega_y = 1.0 - 0.85 \cdot (1 - e^{-5a^b}) \qquad \text{for } \varepsilon_{sy} < \varepsilon_s \leq \varepsilon_{su} \qquad (20.5\text{-}9b)$$

The bond stress–slip curve is considerably influenced by reinforcement yielding, transverse pressure and cracking along the bar, and by cyclic, repeated or sustained loading.

Bond resistance is decreased over the portion of an embedment length once reinforcement yield is reached, Fig. 20.5-3.

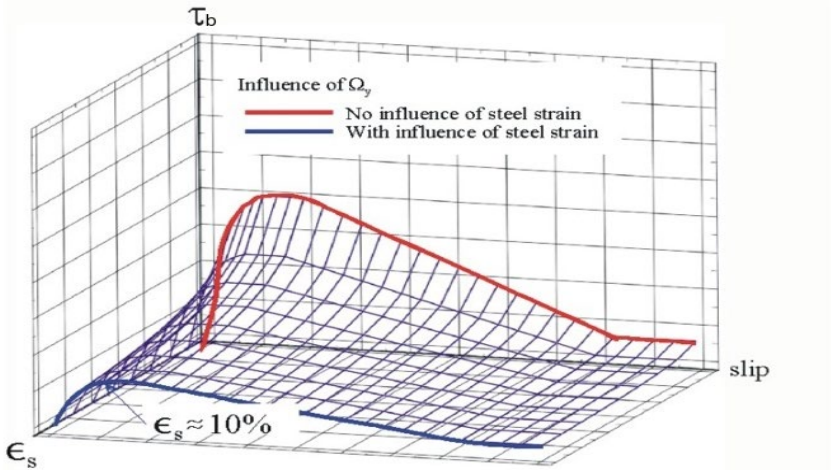


Figure 20.5-4-*: Influence of steel strains on local bond stress-slip relationship on case of pull-out failure. Bond resistance is decreased over the portion of an embedded length once reinforcement yield is reached.*

Bond resistance is decreased in the presence of transverse tension and increased in the presence of compressive stresses perpendicular to the bar axis – Fig. 20.5-4 and Eq. (20.5-11a) and Eq. (20.5-11b).

A negative quantity for p_{tr} in Eq. (20.5-11a) or (20.5-11b) denotes a compressive stress.

with

$$a = \frac{\varepsilon_s - \varepsilon_{sy}}{\varepsilon_{su} - \varepsilon_{sy}} \; ; b = \left[2 - \frac{f_{tm}}{f_{ym}}\right]^2 \qquad (20.5\text{-}10)$$

f_{tm} and f_{ym} are mean values of tensile and yield strength respectively.

20.5.1.3.2 Influence of transverse pressure

If transverse pressure perpendicular to the bar axis is present, the bond stress slip curve for pull-out failure should be modified by the factor $\Omega_{p,tr}$ as:

$$\Omega_{p,tr} = 1.0 - \frac{0.3 p_{tr}}{f_{ctm}} \qquad \text{for } f_{ctm} \geq p_{tr} \geq 0 \text{ MPa (tension)} \qquad (20.5\text{-}11a)$$

$$\Omega_{p,tr} = 1.0 - tanh\left[0.2 \frac{p_{tr}}{0.1 f_{cm}}\right] \qquad \text{for } p_{tr} \leq 0 \text{ MPa (compr.)} \qquad (20.5\text{-}11b)$$

where:

p_{tr} is the mean stress in the concrete (orthogonal to the bar axis) averaged over a volume around the bar with a diameter of 3 Ø.

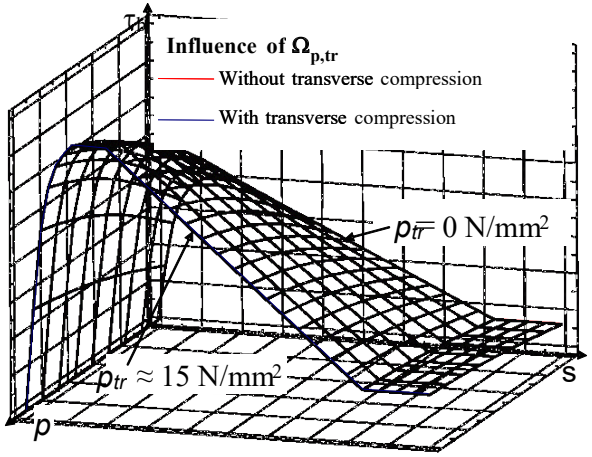


Figure 20.5-5: Influence of transverse pressure on local bond-stress slip relationship in case of pull-out failure.

A transverse tensile stress equal to the tensile strength of the concrete could split the concrete cover and leave zero bond resistance where there is no reinforcement crossing the splitting crack. Eq. (20.5-12) represents behaviour where reinforcement crosses potential splitting cracks and is not stressed above its yield strength.

Eq. (20.5-12) may be considered valid for cracks up to 0.5 mm wide.

Fig. 20.5-6 explains also the calculation of the dissipated energy during monotonic and cyclic loading (area under the bond stress-slip curve). In Fig. 20.5-7 the factor Ω_{cyc} is plotted as a function of the ratio Λ_{cyc}/Λ_0.

20.5.1.3.3 Influence of longitudinal cracking

If cracks parallel to the bar axis and transverse reinforcement are present, the bond stress slip curve for pull-out failure should be modified by the factor Ω_{cr}:

$\Omega_{cr} = 1.0$ where concrete is uncracked parallel to the bar axis

$\Omega_{cr} = 1 - 1.2\ w_{cr}$ where the concrete is cracked parallel to the bar axis and w_{cr} is the crack width [mm]. (20.5-12)

20.5.1.3.4 Influence of reversed cyclic loading

Reversed cyclic loading reduces the bond strength compared to monotonic loading as shown in Fig. 20.5-6.

After the bond stress has reached τ_{bmax} the ordinates (i.e. the stress values) of the bond stress slip curve for pull-out failure should be modified by the factor Ω_{cyc} on subsequent load cycles as:

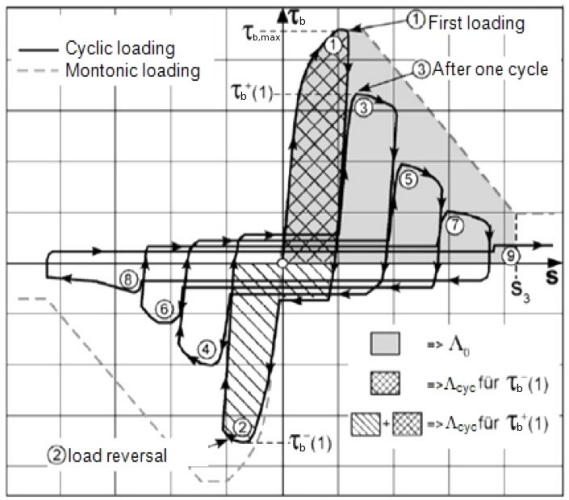


Figure 20.5-6: *Bond-stress slip relationship and definition of the dissipated energy under monotonic and cyclic loading.*

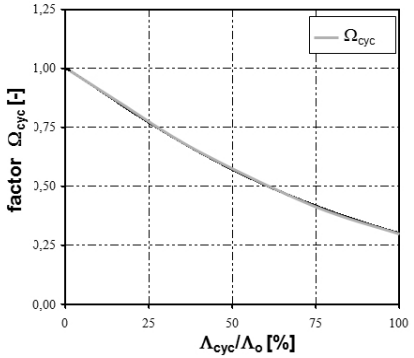


Figure 20.5-7: *Factor Ω_{cyc} as a function of the ratio Λ_{cyc}/Λ_0.*

$$\Omega_{cyc} = e^{\left(-1.2\cdot\left[\frac{\Lambda_{cyc}}{\Lambda_0}\right]^{1.1}\right)} \quad (20.5\text{-}13)$$

where:

Λ_{cyc} dissipated energy during cyclic loading (Fig. 20.5-7);

Λ_0 dissipated energy during monotonic loading (Fig. 20.5-7).

In the absence of test data, the β coefficient for epoxy coated ribbed bars should be taken as 50% of that for a "normal" ribbed bar.

The influence of bond on crack control and tension stiffening may be estimated from the secant modulus of local bond–slip behaviour measured in accordance with RILEM pull-out test procedures at a free end slip of 0.01 mm.

The terms "non-reversed repeated loading" or "fatigue loading" are used here to imply loading in which the bar stress does not change its sign and the bond stress does not exceed τ_{bmax} under short-term monotonic loading. This type of repeated loading produces a progressive increase of slip between steel reinforcement and concrete which may lead to failure at cyclic bond stress levels lower than the ultimate bond strength corresponding to monotonic loading. The rate of increase is influenced principally by the number of load cycles and the load level.

Provided that bond failure does not occur under repeated loading, then the preceding repeated loads do not adversely influence either bond-slip behaviour on subsequent monotonic loading to failure or bond strength at ultimate load.

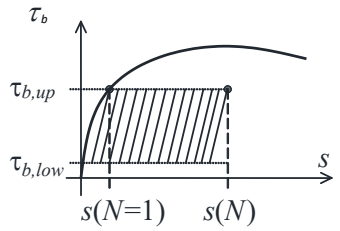


Figure 20.5-8: *Effect of repeated loading on the bond-slip curve*

20.5.1.4 Influence on serviceability

Rules for serviceability behaviour have been derived for ribbed bars with a relative rib area f_r within the range 0.05–0.07. More highly ribbed bars may be capable of developing higher bond stiffness, while coatings may cause a reduction. Coefficients β_{TS} representing the influence of bond in Eqs. (30.5-2), (30.5-11) and (30.5-12) and β in Eq. (30.5-25) in Section 30.5.2 may be modified in proportion to measured bond stiffness.

20.5.1.5 Influence of fatigue loading

Non-reversed repeated loading increases the relative slip between concrete and ribbed reinforcement with load cycles, Fig. 20.5-8. The increase of slip $s(N)$ after N load cycles can be calculated according to the following equation:

$$s(N) = s(N=1)N^b \qquad (20.5\text{-}14)$$

where $N = 1$ means quasi-static monotonic loading and factor b takes account of the load level as follows:

$$b = \begin{cases} 0.107 & ,\tau_{b,up}/\tau_{bu} \le 0.51 \\ 0.45\left(\tau_{b,up}/\tau_{bu}\right)^{2.13} & ,\tau_{b,up}/\tau_{bu} > 0.51 \end{cases} \qquad (20.5\text{-}15)$$

In the above equation, $\tau_{b,up}$ is the bond stress under maximum load level and τ_{bu} = min (τ_{bmax}, $\tau_{bu,split}$) is the monotonic bond strength under quasi-static loading

Eq. (20.5-16) has been derived for short pull-out test conditions, where the bond stress distribution along the embedded length can be considered uniform. Eq. (20.5-16) is obtained by considering that fatigue failure initiates when the relative slip $s(N)$ reaches the peak s_1 referred to in Table 20.5-1.

For further information on bond under cyclic loading, see chapter 3, "Bond under repeated loading", of *fib* Bulletin 10 [20-21].

During the last few years, new types of concrete came into the market, including Fibre Reinforced Concrete (FRC), concrete with conventional aggregates, polymer concrete, etc.

Therefore, the rules proposed in section 20.2 and 20.3 cannot be applied to new types of concrete or different types of rebars.

In order to verify the applicability of the rules or the need of correction factors, a standard Bond Test is proposed in section 20.6.

Based on available literature the bond performance between steel reinforcement and LW (Low weight) concrete or RSFRC (Recycled Steel Fiber Reinforced Concrete) are linked to several parameters that need to be investigate with specific experimental tests (Section 20.6).

Only the pullout mode needs to be considered for plain surface bars. Many of the general points of principle for local bond modelling detailed in 20.5.1.1 are also applicable to plain surface bars and are not repeated here.

The parameters given in Table 20.5-2 are mean values. The scatter in measured slip is considerable, especially for small values of slip. For a given value of the slip the coefficient of variation of the bond stresses may amount to □pprox.. 30%. The

according to Table 20.5-1. Fatigue failure can occur if the number of applied load cycles (N) is sufficiently high to reach the fatigue resistance (N_f). For constant amplitude load cycles, bond fatigue strength can be evaluated with the following equation:

$$N_f = \left(\frac{\tau_{b,up}}{\tau_{bu}} \right)^{\frac{-1}{\alpha \cdot b}} \qquad (20.5\text{-}16)$$

where b is defined in Eq. (20.5-15) and α is the exponential factor defining the first branch of the quasi-static bond-slip curve (α = 0.4 can be used according to Table 20.5-1).

20.5.2 Local bond-slip relationship of steel reinforcement in non-conventional concretes

Anchorage and lap behaviour equations described in Section 20.2 are based on a wide database including worldwide experiments carried out on normal concrete on different types of rebars. Equations proposed in Section 20.2 are valid within a specific range of material properties.

20.5.3 Local bond-slip for plain (non-ribbed) surface bars

For monotonic loading the value τ_b of the bond stresses between concrete and reinforcing bar for pull-out failure can be calculated as a function of the relative displacement s parallel to the bar axis according to Eq. 20.5-17.

$$\tau_b = \tau_{bmax}\, Min\left[\left({}^{s}/_{s_1} \right)^{\alpha_r}, \left({}^{s}/_{s_1} \right)^{\alpha_f} \right] \qquad (20.5\text{-}17)$$

scatter is due to the use of different test specimens, the differences in the state of stress in the concrete surrounding the reinforcing bar, the different measuring techniques as well as to the different loading and deformation velocities. The heterogeneity of the concrete and the surface characteristics of the reinforcing bars (initial surface condition, whether millscale, surface corrosion etc., diameter) also have a significant influence on the τ - s relationship.

The parameters given in Table 20.5-2 are valid for plain (i.e. non-ribbed and non-twisted) reinforcing steel, depending on the main influencing factors: roughness of the bar surface, bond conditions and concrete strength. They are valid for confined concrete and are applicable only in loading states with concrete not subjected to lateral tension, in the elastic range of the reinforcement and for those parts of the reinforcing bar with the distance $x > 2Ø$ from a transverse crack. For those parts of the reinforcing bars with a distance $x \leq 2Ø$ from a transverse crack, the bond stress τ_b and the slip s are to be reduced by the factor λ according to Eq. (20.5-7).

Table 20.5-2 *Parameters defining the bond stress–slip relationship of plain surface bars in well confined conditions.*

	Hot rolled bars		Cold drawn bars and wires	
	Good bond conditions	All other bond conditions	Good bond conditions	All other bond conditions
s_1	0.25 mm	0.25mm	0.02 mm	0.02mm
α_r	0.2	0.2	0.2	0.2
α_f	-0.2	-0.2	-0.2	-0.2
η_4	$5.0(25/Ø)^{0.2}$	$2.5(25/Ø)^{0.2}$	$2.5(25/Ø)^{0.2}$	$1.25(25/Ø)^{0.2}$
τ_{bmax}	$\eta_4\sqrt{(f_{cm}/25)}$			

20.5.4 Local bond-slip under elevated temperature

Exposure to elevated temperatures leads to a degradation in the material properties of concrete such as the tensile strength, compressive strength and E-modulus. Consequently, the bond strength between reinforcement and concrete also degrades. The expression given in Eq. (20.5-18) provides a conservative estimate of the degradation in bond strength with elevated temperature (see Fig. 20.5-7). The expression is valid for both pullout failure and splitting failure and reaches a value of zero at 740°C.

For ribbed reinforcing bars, the degradation of bond strength between ribbed reinforcing bars and concrete due to the exposure to elevated temperature should be considered as:

$$\tau_{bmax,T} = \tau_{b\max,20} \cdot \left(1 - \frac{T-20}{720}\right) \text{ for } 20°C < T \leq 740°C \qquad (20.5\text{-}18)$$

where,

$\tau_{bmax,T}$ is the bond strength corresponding to an elevated temperature T (°C)

$\tau_{bmax,20}$ is the bond splitting strength according to Eq. (20.5-5) or the bond pullout strength given by Eqs. (20.5-1)–(20.5-4), whichever is smaller under ambient conditions.

T is the temperature, measured at the level of the reinforcing bar.

The bond stress-slip relationship between deformed reinforcement bar and concrete subjected to an elevated temperature T is given as:

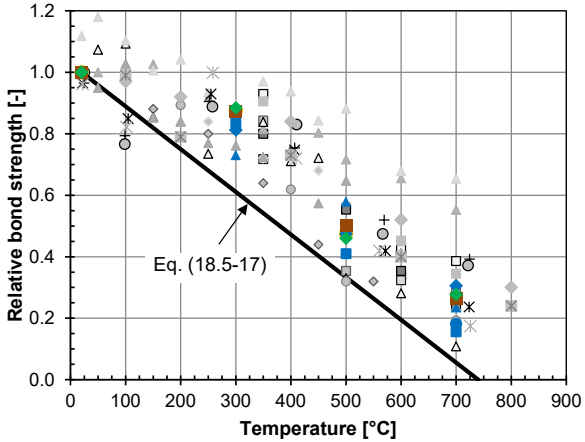


Figure 20.5-8: *Experimental evidence of bond strength degradation at elevated temperature in quasi-static thermal conditions and comparison with Eq. (20.5-18).*

Tests have shown that if the applied bond stress on the reinforcement under ambient conditions is less than 60% of its ambient bond strength, the influence of load on bond capacity at elevated temperatures is negligible. Since the design bond strength is typically approx. 50% of its mean bond strength, this influence is ignored in Eq. (20.5-18).

Exposure to elevated temperatures lead to a reduction in the bond stiffness. Evaluation of the test data shows that the degradation of secant stiffness corresponding to the first attainment of peak in case of can be considered as

(a) Bond pull-out failure

$$k_T = k_0\left(1-\frac{T-20}{5000}\right) \quad \text{for } T \le 300°C \qquad (20.5\text{-}26)$$

$$k_T = k_0\left(0.94-\frac{T-300}{600}\right) \quad \text{for } T > 300°C \qquad (20.5\text{-}27)$$

(b) Bond splitting failure

$$k_T = k_0\left(1-\frac{T-20}{800}\right) \qquad (20.5\text{-}28)$$

where:

k_T is the bond stiffness corresponding to an elevated temperature T (°C)

k_0 is the secant stiffness corresponding to the first attainment of peak bond stress–slip defined by Eq. (20.5-1)–(20.5-4).

Temperature, T is measured at the level of the reinforcing bar.

$$\tau_{bT} = \tau_{bmax,T}\,(s/s_{1,T})^{\alpha} \quad \text{for } 0 \le s \le s_{1,T} \qquad (20.5\text{-}19)$$

$$\tau_b = \tau_{bmax,T} \quad \text{for } s_{1,T} \le s \le s_{2,T} \qquad (20.5\text{-}20)$$

$$\tau_b = \tau_{bmax,T} - (\tau_{bmax,T} - \tau_{bf,T})\,(s\text{-}s_{2,T})/(s_{3,T}\text{-}s_{2,T}) \quad \text{for } s_{2,T} \le s \le s_{3,T} \qquad (20.5\text{-}21)$$

$$\tau_b = \tau_{bf,T} \quad \text{for } s_{3,T} < s \qquad (20.5\text{-}22)$$

where, $\tau_{bmax,T}$ is given by Eq. (20.5-18). The values of $s_{1,T}$, $s_{2,T}$ and $s_{3,T}$ can be obtained as:

(a) For bond pullout failure

$$s_{1T} = 6.94 s_{1,20}\left(\frac{740-T}{5020-T}\right) \text{ for } 20°C < T \le 300°C \qquad (20.5\text{-}23)$$

$$s_{1T} = 0.83 s_{1,20}\left(\frac{740-T}{864-T}\right) \text{ for } 300°C < T \le 740°C \qquad (20.5\text{-}24)$$

where:

$s_{1,20}$ is the displacement s_1 under ambient conditions according to Fig. 20.5-1 and Table 20.5-1.

$s_{2,T} = 2s_{1,T}$

$s_{3,T} = c_{clear}$ = the clear distance between ribs

(b) For bond splitting failure

$$s_{1T} = 1.11 s_{1,20}\left(\frac{740-T}{820-T}\right) \qquad (20.5\text{-}25)$$

where:

$s_{1,20}$ is the displacement s_1 under ambient conditions for splitting failure mode according to Table 20.5-1.

$s_{2,T} = s_{1,T}$

$s_{3,T} = 1.2\ s_{1,T}$ for no stirrups and $0.5c_{clear}$ for with stirrups case

20.5.5 Local bond-slip under fire

After a defined fire exposure duration (in minutes), the bond capacity, $\tau_{bu,fi}$ of a deformed reinforcing bar can be calculated as:

$$\tau_{bu,fi} = k_{fi}\ \tau_{bu,0} \quad (20.5\text{-}29)$$

The values of the coefficient k_{fi} depend on the time of fire exposure and shall be read from Table 20.5-3.

The basic value of bond strength under ambient conditions, $\tau_{bu,0}$ is given by Eq. (20.5.5).

The values of slip of deformed reinforcing bar, $s_{i,fi}$ after an exposure to fire for a defined exposure duration can be obtained by using Eq. (20.5.30).

$$s_{i,fi} = k_{s,fi}\ s_{i,20} \quad (20.5\text{-}30)$$

where:

$s_{i,20}$ is the value of slip (s_1, s_2 or s_3) under ambient conditions, to be read from Table 20.5-1. The recommended values for the slip modification factor $k_{s,fi}$ shown in Fig. 20.5-9.

The values of the coefficient k_{fi} shall be read from the Table 20.5-3.

Table 20.5-3. *The values of the coefficient* k_{fi}

Exposure duration [min]	**Value of k_{fi}**
0 min	1.0
15 min	0.5
30 min	0.4
60 min	0.33
90 min	0.30

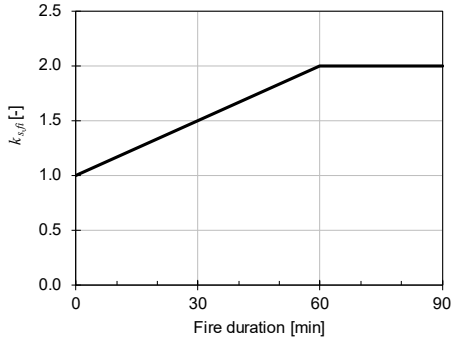


Figure 20.5-9: *Slip modification factor* $k_{s,fi}$ *for bond slip values*

20.6 Standard test method for anchorages

The test is intended to verify that existing provisions for anchorage and laps in the *fib* Model Code when non-conventional concretes and rebars are adopted (i.e. fibre reinforced concrete, recycled aggregate concrete, etc.). However, it could also be useful for testing bond characteristics of novel reinforcement.

This test method describes procedures to assess the bond strength (anchorage capacity) of steel reinforcing bars in concrete.

This test method is intended to determine the effects of concrete cover, anchorage length, transverse reinforcement, concrete strength, casting position on the bond strength of deformed steel reinforcing bars having a diameter of 16 mm. The test method allows a pressure transverse to the potential splitting plane to be applied.